



General Chemistry I

Recitation 5

Siyuan Chen

Oct. 27

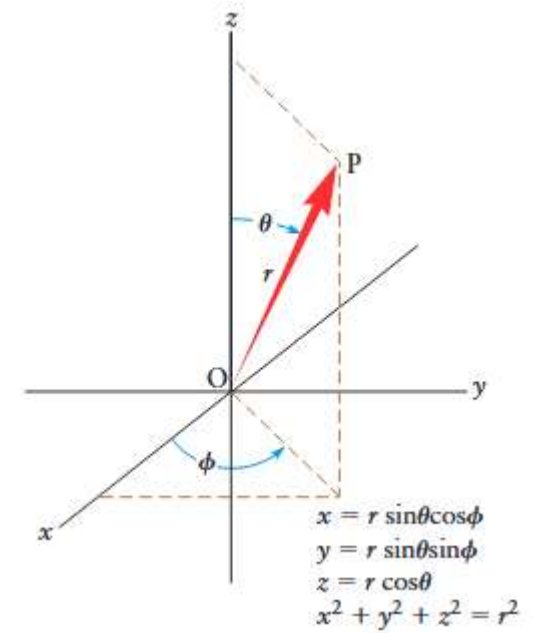
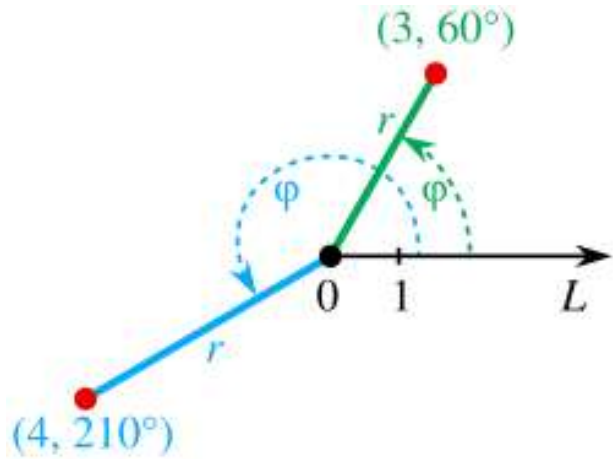
Notice !

- Time: Oct. 29 (Wednesday) 90minutes
- Key Points:
 - Chemical Formulas & Nomenclatures
 - Significant Figures
 - Chemical Calculation Involving Sigfigs
 - Lewis Diagrams & Formal Charge
 - Molecular Structure
 - Bohr Model
 - Quantum Mechanics for Hydrogen Atom (Single-electron atom system)
 - Wave Function and Atomic Orbitals
- Preparation: Student ID card, calculator, stationeries(pen, pencil etc.)
- Please attend the midterm ON TIME!!!

Tips: Please get familiar with your calculator.

Wave functions

From polar coordinate system to spherical coordinate system



$$(x, y) \xrightarrow{\text{Transformation}} (r, \varphi)$$

$$(x, y, z) \xrightarrow{\text{Transformation}} (r, \theta, \varphi)$$

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_{lm}(\theta, \phi)$$

radial part angular part

Basis and Space

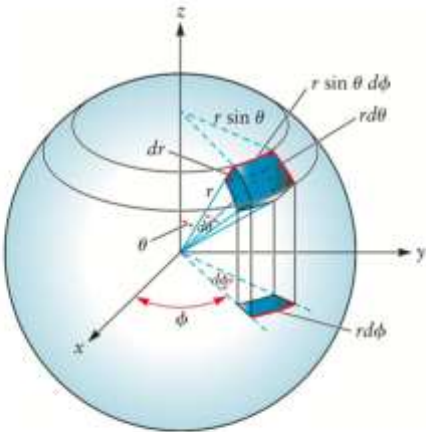
(Review your LINEAR ALGEBRA)

Wave functions

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

radial part angular part

$$(\psi_{n\ell m})^2 dV = [R_{n\ell}(r)]^2 [Y_{\ell m}(\theta, \phi)]^2 dV$$



$$dV = dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

Write the wave functions including radial and angular part on your own:

1s 2s 2p

TABLE 5.2 Angular and Radial Parts of Wave Functions for One-Electron Atoms

Angular Part $Y(\theta, \phi)$

Radial Part $R_{n\ell}(r)$

$$\ell = 0 \left\{ Y_s = \left(\frac{1}{4\pi} \right)^{1/2} \right.$$

$$R_{1s} = 2 \left(\frac{Z}{a_0} \right)^{3/2} \exp(-\sigma)$$

$$R_{2s} = \frac{1}{2\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) \exp(-\sigma/2)$$

$$R_{3s} = \frac{2}{81\sqrt{3}} \left(\frac{Z}{a_0} \right)^{3/2} (27 - 18\sigma + 2\sigma^2) \exp(-\sigma/3)$$

$$\ell = 1 \left\{ \begin{aligned} Y_{p_x} &= \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \cos \phi \\ Y_{p_y} &= \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \sin \phi \\ Y_{p_z} &= \left(\frac{3}{4\pi} \right)^{1/2} \cos \theta \end{aligned} \right.$$

$$R_{2p} = \frac{1}{2\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma \exp(-\sigma/2)$$

$$R_{3p} = \frac{4}{81\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} (6\sigma - \sigma^2) \exp(-\sigma/3)$$

$$\ell = 2 \left\{ \begin{aligned} Y_{d_{z^2}} &= \left(\frac{5}{16\pi} \right)^{1/2} (3 \cos^2 \theta - 1) \\ Y_{d_{xz}} &= \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \cos \phi \\ Y_{d_{yz}} &= \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \sin \phi \\ Y_{d_{xy}} &= \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \sin 2\phi \\ Y_{d_{x^2-y^2}} &= \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \cos 2\phi \end{aligned} \right.$$

$$R_{3d} = \frac{4}{81\sqrt{30}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 \exp(-\sigma/3)$$

$$\sigma = \frac{Zr}{a_0}$$

$$a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10} \text{ m}$$

Wave functions

$$\left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \right] \psi = E\psi$$

Solutions of the Schrödinger equation for the one-electron atom exist only for **particular values of the energy**

$$E = E_n = -\frac{Z^2 e^4 m_e}{8\epsilon_0^2 n^2 \hbar^2} \quad n = 1, 2, 3, \dots$$

$$E_n = -\frac{Z^2}{n^2} \quad (\text{rydberg}) \quad n = 1, 2, 3, \dots$$

The energy of a one-electron atom depends **only on the principal quantum number n**, because the potential energy depends only on the radial distance.

The integer **n** (**principal quantum number**, 主量子数) indexes the individual energy levels. These are identical to the energy levels predicted by the Bohr theory.

The Schrödinger equation **also quantizes L^2** , the square magnitude of the angular momentum (**L**), as well as **L_z** , the projection of the angular momentum along the z-axis (**L_z**).

$$L^2 = \ell(\ell + 1) \frac{\hbar^2}{4\pi^2} \quad \ell = 0, 1, \dots, n-1$$

The **angular momentum quantum number ℓ** (角量子数), may take on any integral value from 0 to n-1.

$$L_z = m \frac{\hbar}{2\pi} \quad m = -\ell, -\ell + 1, \dots, 0, \dots, \ell - 1, \ell$$

The **magnetic quantum number m** (磁量子数), may take on any integral value from - ℓ to ℓ . Its value governs the behavior of the atom in an external magnetic field.

$$n = 1, 2, 3, \dots$$

$$\ell = 0, 1, \dots, n-1$$

$$m = -\ell, -\ell + 1, \dots, 0, \dots, \ell - 1, \ell$$

When n=1. Only one set of quantum states: n=1, l=0, m=0

When n=2. Four sets of quantum states: (2, 0, 0); (2, 1, -1); (2, 1, 0); (2, 1, 1)

When n=n. $1+3+5+\dots+(2n-1)=n^2$ sets of quantum states

Each set (**n, ℓ , m**) identifies a specific **quantum state** of the atom in which the electron has energy equal to E_n .

When $n > 1$, a total of n^2 specific quantum states correspond to the single energy level E_n ; consequently, this set of states is said to be **degenerate** (简并).

Wave functions

$\ell = 0$ with s, $\ell = 1$ with p, $\ell = 2$ with d, $\ell = 3$ with f, $\ell = 4$ with g

1s

2s 2p

3s 3p 3d

4s 4p 4d 4f

TABLE 5.1 Allowed Values of Quantum Numbers for One-Electron Atoms

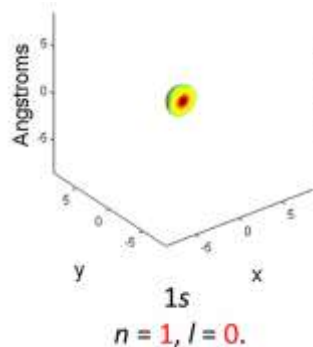
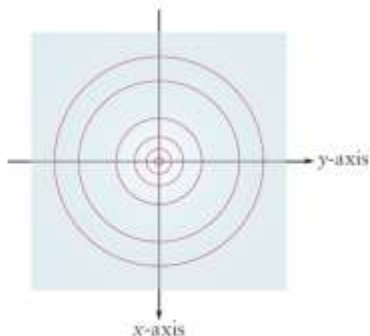
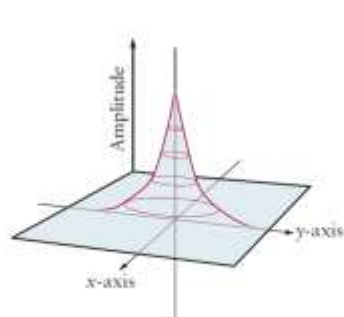
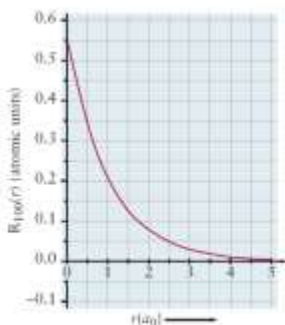
n	1	2		3		
ℓ	0	0	1	0	1	2
m	0	0	$-1, 0, +1$	0	$-1, 0, +1$	$-2, -1, 0, +1, +2$
Number of degenerate states for each ℓ	1	1	3	1	3	5
Number of degenerate states for each n	1	4		9		

Wave functions

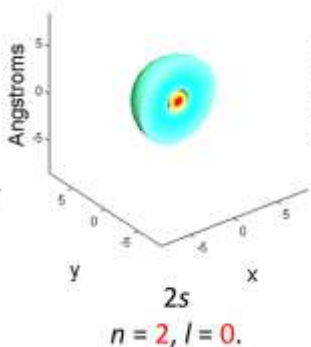
Functions, nodes and orbitals

The hydrogen atom: 1s orbitals

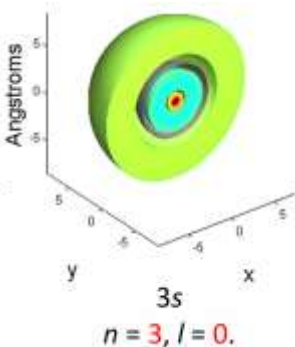
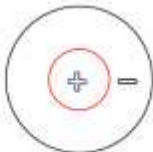
$$\psi_{1s} = \frac{1}{\sqrt{\pi}} a_0^{-3/2} \cdot \exp\left(-\frac{r}{a_0}\right)$$



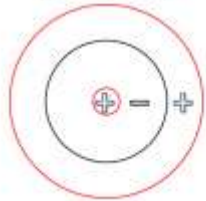
1s
 $n = 1, l = 0.$



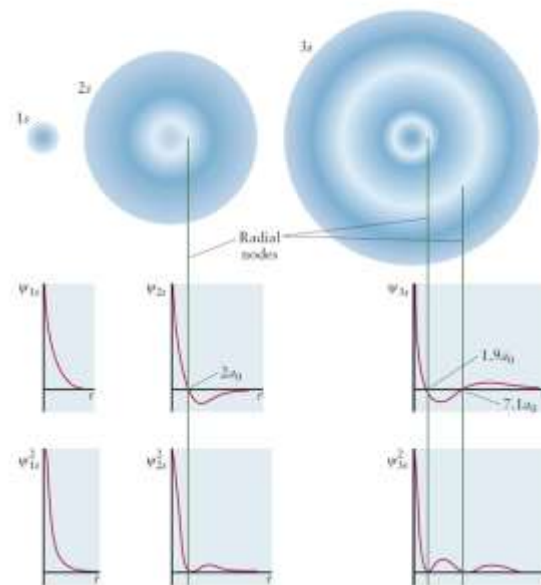
2s
 $n = 2, l = 0.$



3s
 $n = 3, l = 0.$



The hydrogen atom: s orbitals



$$R_{1s} = 2 \left(\frac{Z}{a_0} \right)^{3/2} \exp(-\sigma)$$

$$R_{2s} = \frac{1}{2\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) \exp(-\sigma/2)$$

$$R_{3s} = \frac{2}{81\sqrt{3}} \left(\frac{Z}{a_0} \right)^{3/2} (27 - 18\sigma + 2\sigma^2) \exp(-\sigma/3)$$

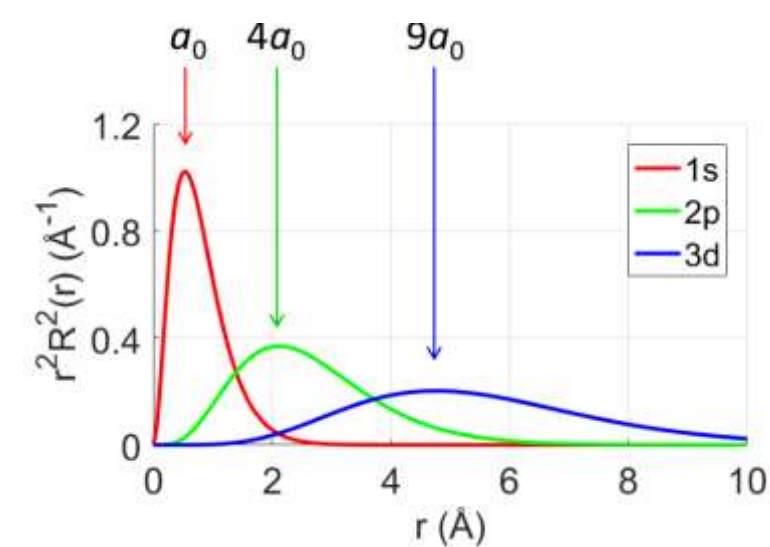
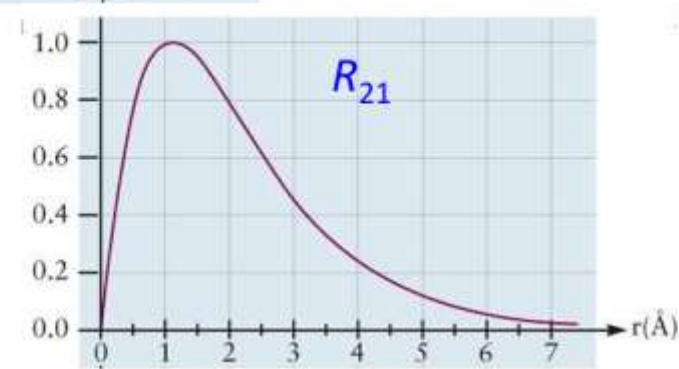
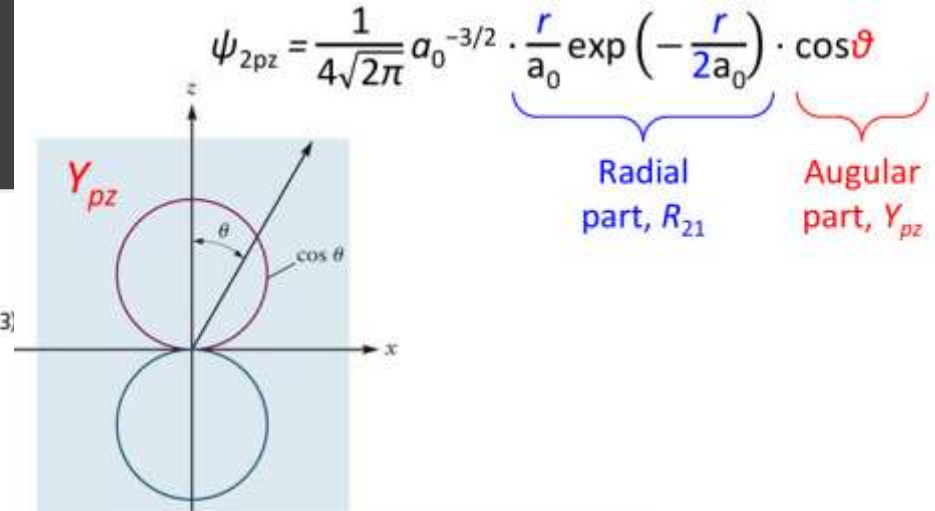
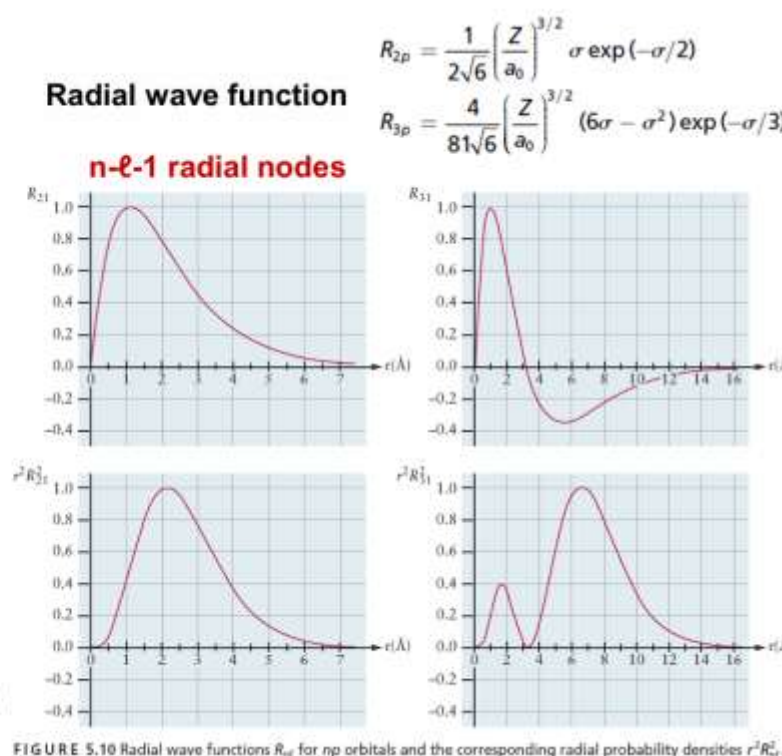
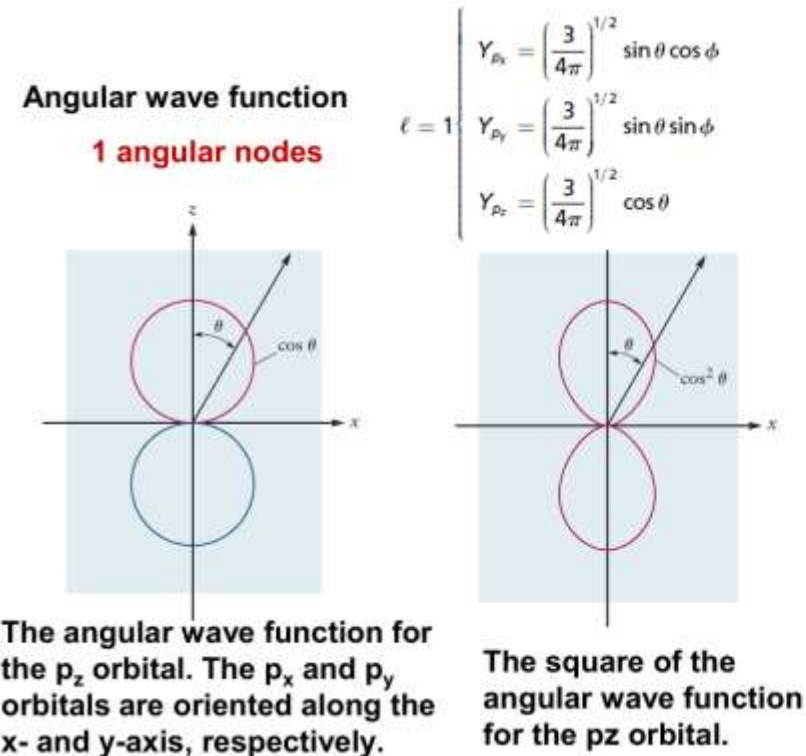
函数的性态如何？

几个节点？

节点是由径向部分（radial part）还是角向部分（angular part）得到？
对应怎样的电子云？

Is that “positive” charge?

Wave functions



What does $r^2 R^2$ represent?

函数的性态如何？

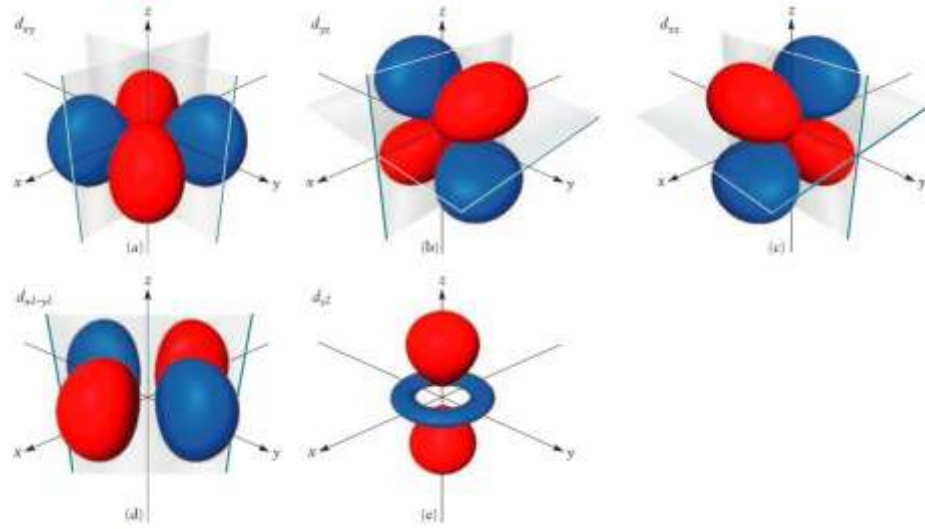
几个节点？

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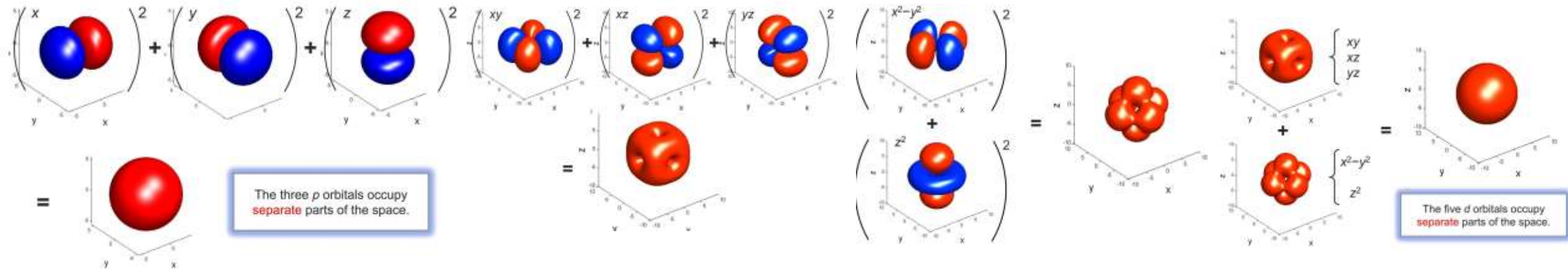
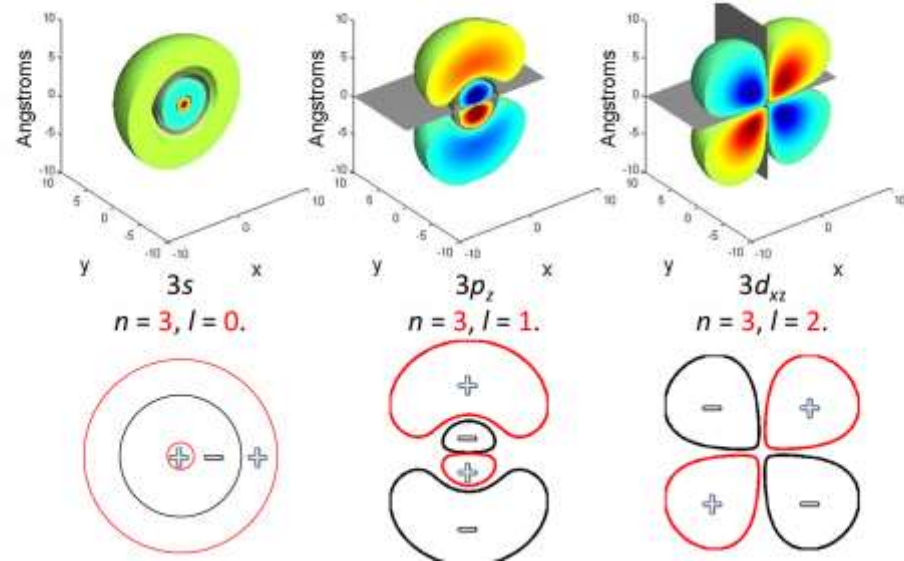
对应怎样的电子云？

Wave functions

d orbital: 2 angular nodes, n-3 radial nodes. n-1 total nodes



The hydrogen atom: Comparison of 3s, 3p, 3d orbitals



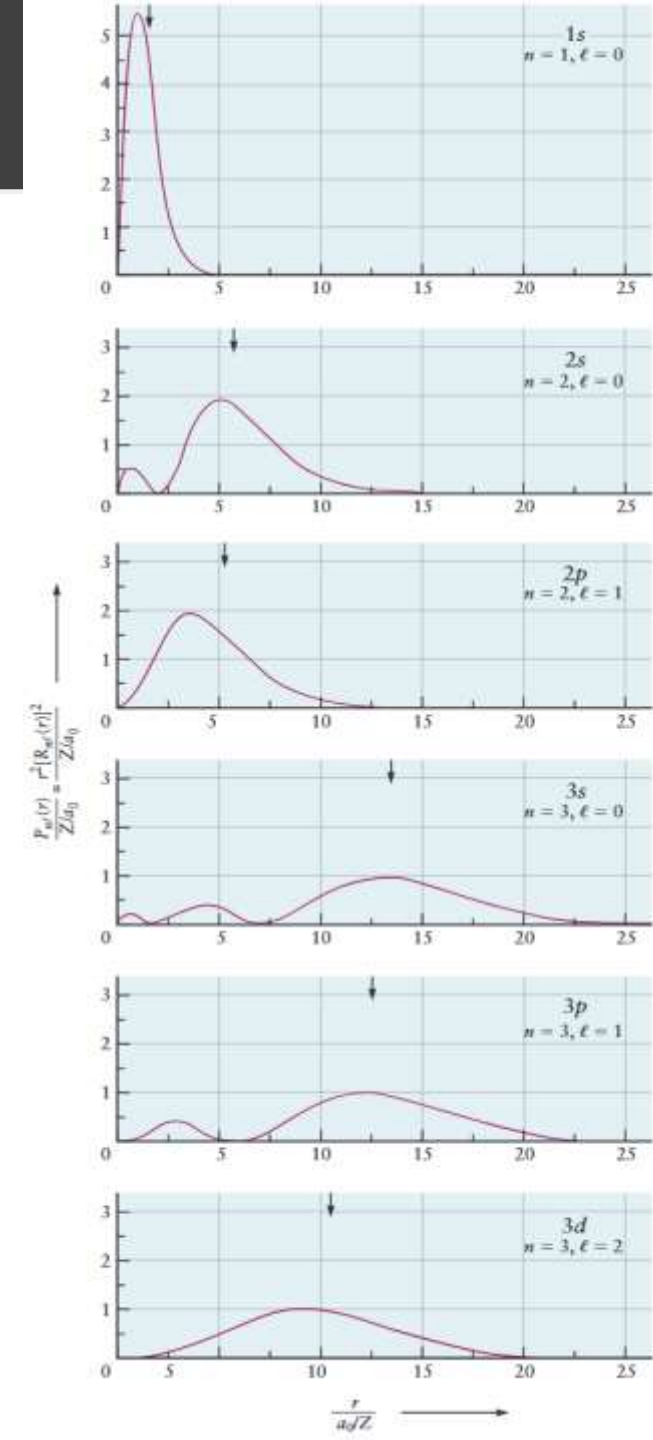
Linear combination of functions !

Wave functions

We summarize the important features of orbital shapes and sizes as follows:

1. For a given value of ℓ , an increase in n leads to an increase in the average distance of the electron from the nucleus, and therefore in the size of the orbital (see Figs. 5.8 and 5.10).
2. An orbital with quantum numbers n and ℓ has ℓ angular nodes and $n - \ell - 1$ radial nodes, giving a total of $n - 1$ nodes. An angular node is defined by a plane. A radial node is defined by a spherical surface. For a one-electron atom or ion, the energy depends only on the number of nodes—that is, on n but not on ℓ or m . The energy increases as the number of nodes increases.
3. As r approaches 0, $\psi(r, \theta, \phi)$ vanishes for all orbitals except s orbitals; thus, only an electron in an s orbital can “penetrate to the nucleus,” that is, have a finite probability of being found right at the nucleus.

- Orbitals:
- s orbitals: radial node (nodal sphere)
- p orbitals: $1 \times$ angular node (nodal plane)
- d orbitals: $2 \times$ angular node (nodal plane)



Review—— Quiz

1. Label the orbitals described by each of the following sets of quantum numbers: (a) $n=3, l=2$ (b) $n=7, l=4$ (c) $n=5, l=1$

2. How many radial nodes and how many angular nodes does each of the orbitals in Problem 1 have?

统计数据		状态分布		成绩分布	
计数	63	空	2	大于 100	0
最小值	0	进行中	0	90 - 100	54
最大值	10.00	需要评分	0	80 - 89	0
范围	10.00	免除	0	70 - 79	1
平均值	9.03			60 - 69	1
中位数	10.00			50 - 59	4
标准偏差	2.42			40 - 49	0
方差	5.87			30 - 39	0
				20 - 29	0
				10 - 19	0
				0 - 9	3
				小于 0	0

Review

Wave function analysis

- **2.5** (0.6 point) The wave function for the H 3s orbital is

$$\psi_{3s}(r) = \frac{1}{81\sqrt{3}\pi} a_0^{-3/2} (27 - 18\sigma + 2\sigma^2) \exp(-\sigma/3), \quad \sigma = r / a_0,$$

- where r is the distance between the electron and the nucleus, and $a_0 = 0.53 \text{ \AA}$ is Bohr radius. Find the values of r (in $\text{\AA}$) for which nodes exist for the H 3s orbital.

Review

- **2.4** (1.2 point) The wave function of an electron in the lowest (that is, ground) state of the hydrogen atom is

$$\psi(r) = \left(\frac{1}{\pi a_0^3}\right)^{1/2} \exp\left(-\frac{r}{a_0}\right)$$

$$a_0 = 0.529 \times 10^{-10} \text{ m}$$

- (a) What is the probability of finding the electron inside a sphere of volume 1.0 pm^3 , centered at the nucleus ($1 \text{ pm} = 10^{-12} \text{ m}$)?
- (b) What is the probability of finding the electron in a volume of 1.0 pm^3 at a distance of 52.9 pm from the nucleus, in a fixed but arbitrary direction?

$$\int_{-\infty}^{+\infty} P(x) dx = \int_{-\infty}^{+\infty} [\psi(x)]^2 dx = 1$$

Review

Compare the $3p$ and $4d$ orbitals of a hydrogen atom with respect to the (a) number of radial and angular nodes and (b) energy of the corresponding atom.

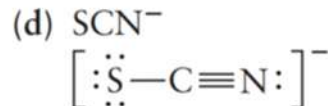
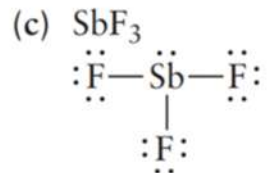
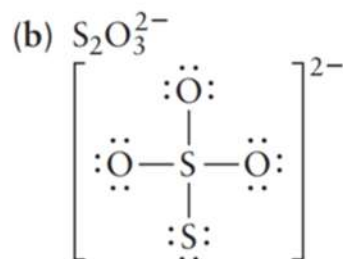
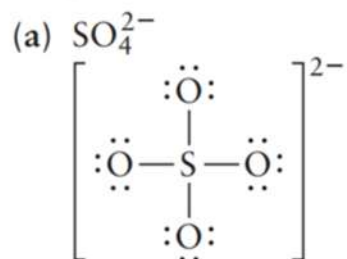
Solution

- (a) The $3p$ orbital has a total of $n - 1 = 3 - 1 = 2$ nodes. Of these, one is angular ($\ell = 1$) and one is radial. The $4d$ orbital has $4 - 1 = 3$ nodes. Of these, two are angular ($\ell = 2$) and one is radial.
- (b) The energy of a one-electron atom depends only on n . The energy of an atom with an electron in a $4d$ orbital is higher than that of an atom with an electron in a $3p$ orbital, because $4 > 3$.

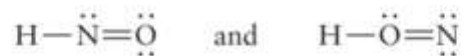
Review

Lewis diagrams...Again

39. Assign formal charges to all atoms in the following Lewis diagrams.

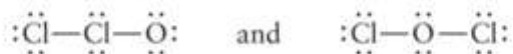


41. Determine the formal charges on all the atoms in the following Lewis diagrams.



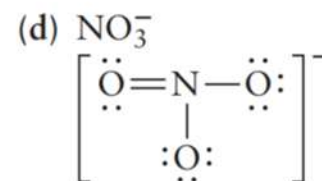
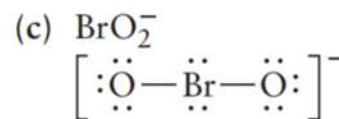
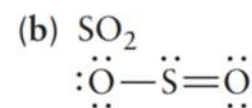
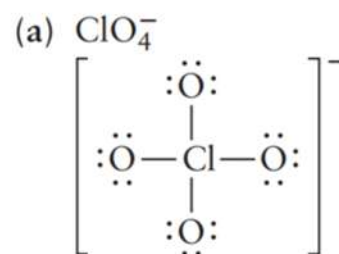
Which one would best represent bonding in the molecule HNO?

42. Determine the formal charges on all the atoms in the following Lewis diagrams.

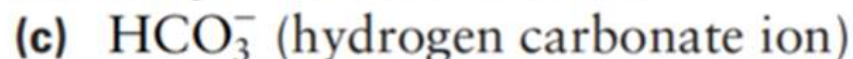


Which one would best represent bonding in the molecule Cl_2O ?

40. Assign formal charges to all atoms in the following Lewis diagrams.



51. Draw Lewis electron dot diagrams for the following species, indicating formal charges and resonance diagrams where applicable.



Review

57. Draw Lewis diagrams for the following compounds. In the formula the symbol of the central atom is given first. (*Hint:* The valence octet may be expanded for the central atom.)

(a) PF_5 (b) SF_4 (c) XeO_2F_2

58. Draw Lewis diagrams for the following ions. In the formula the symbol of the central atom is given first. (*Hint:* The valence octet may be expanded for the central atom.)

(a) BrO_4^- (b) PCl_6^- (c) XeF_6^+

3. Draw Lewis electron dot diagrams for the following species, indicating formal charges and resonance diagrams where applicable.

(a) HNC (central N atom)

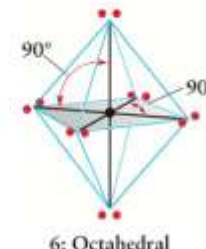
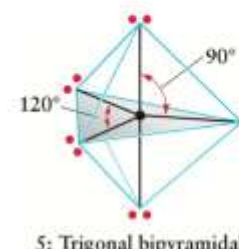
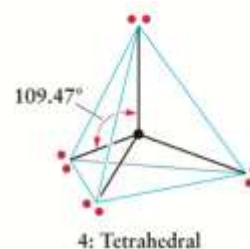
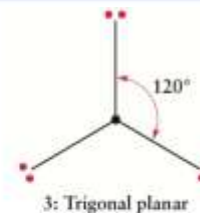
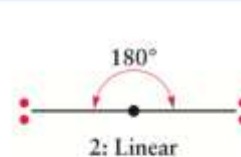
(b) SCN^- (thiocyanate ion)

(c) H_2CNN (the first N atom is bonded to the carbon and the second N)

4. For each of the following molecules or molecular ions, give the steric number and sketch and name the approximate molecular geometry. In each case, the central atom is listed first and the other atoms are all bonded directly to it.

(a) PF_3 (b) SO_2Cl_2 (c) PF_6^- (d) ClO_2^- (e) GeH_4

SN of A	Molecular type X is a bonded atom; E is a lone pair	Predicted shape
2	AX_2	Linear (线形)
3	AX_3	Trigonal planar (三角形)
	AX_2E	Bent (弯曲形)
4	AX_4	Tetrahedral (四面体)
	AX_3E	Trigonal pyramidal (三角锥)
	AX_2E_2	Bent
5	AX_5	Trigonal bipyramid (三角双锥)
	AX_4E	Distorted see-saw (跷跷板)
	AX_3E_2	Distorted T (T字)
	AX_2E_3	Linear
6	AX_6	Octahedral (八面体)
	AX_5E	Square pyramidal (四角锥)
	AX_4E_2	Square planar (正方平面)



Review

Quantum mechanics and Bohr's model

Express the velocity of the electron in the Bohr model for fundamental constants (m_e , e , h , ϵ_0), the nuclear charge Z , and the quantum number n . Evaluate the velocity of an electron in the ground states of He^+ ion and U^{91+} . Compare these velocities with the speed of c . As the velocity of an object approaches the speed of light, relativistic effects become . In which kinds of atoms do you expect relativistic effects to be greatest?

$$v = \frac{nh}{2\pi m_e r} \quad \text{and} \quad r = \frac{\epsilon_0 n^2 h^2}{Z \pi e^2 m_e}.$$

Although the Bohr model does not apply to atoms or ions with more than one electron, it works reasonably well for the $n = 2$ to $n = 1$ transition (denoted $K\alpha$ in physics) for most metal elements.

- (a) Derive the formula for calculating the $K\alpha$ emission wavelength from the atomic number Z and physical constants.
- (b) Physicist Henry Moseley discovered in 1913 that the experimental $K\alpha$ wavelength depends on $Z - 1$ instead of Z . What will be the metal that emits a $K\alpha$ X-ray of $1.54 \text{ \AA}$ wavelength?

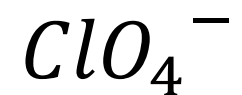
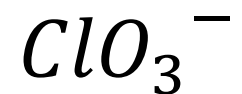
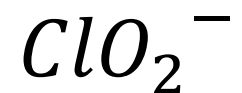
Review

Formula and Nomenclature

TABLE 3.10

Prefixes Used for Naming
Binary Covalent Compounds

Number	Prefix
1	<i>mono-</i>
2	<i>di-</i>
3	<i>tri-</i>
4	<i>tetra-</i>
5	<i>penta-</i>
6	<i>hexa-</i>
7	<i>hepta-</i>
8	<i>octa-</i>
9	<i>nona-</i>
10	<i>deca-</i>



Some advice for Midterm review

1. Please carefully read the textbook and course slides (you could use translators or applications like ZOTERO if you have difficulties in reading English texts)
2. Get familiar with the formulas
3. Homework and quizzes are good materials for your review
4. You can ask instructors for help during Office Hour if you still have problems
5. Do some extra exercises if you need